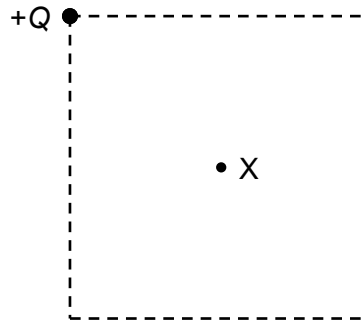


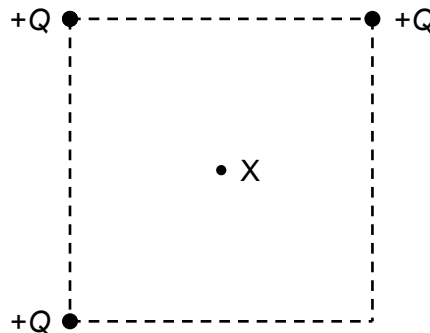
downward force in downward field (force and field same direction) hence positive

21 A point charge $+Q$ is placed at the top-left hand corner of a square as shown.



At the centre X of the square, the electric field strength is E and the electric potential is V .

Two additional point charges of the same magnitude $+Q$ are placed at the top-right and bottom-left corners of the square as shown.



What are the electric field strength and the electric potential at the centre X of the square?

		electric field strength	electric potential	
	A	E	$1.5V$	
	B	E	$3V$	
	C	$2E$	$3V$	
	D	$3E$	$3V$	

Ans: B

The field strength due to the top right corner and bottom left corner cancels out.

Hence the resultant field strength is E .

electric potential = $3V$ (electric potential adds up and all charges at same distance from X)

22 A graph of current against potential difference for a component is given below.